

Live parametric CAD panel (v0.81.0)

The 3-D models were already procedural; now you can drive the parameters and the geometry rebuilds live.

📄 Edit parameters

- Family selector + Length / Width / Height /
- Diameter / Bore — plus Teeth (gears) and
- Coils/turns (springs).
- Cited from FLIS, but editable.

Real parametric inputs.

⚙️ Procedural rebuild

- partgeo.js generates the mesh from the params
- (hex head + threads, races+balls, gear teeth,
- helical coils, holed gaskets...). Edits rebuild
- the geometry live — WebGL or SVG fallback.

Generated, not a fixed model.

🔗 Cited + reset

- Dimensions come from FLIS characteristics.
- 🔗 Reset to FLIS dims restores the cited values.
- 13 families, all verified valid; teeth/turns
- measurably change the geometry.

Explore, then snap back.

What this is (and isn't)

This is procedural / parametric CAD: the geometry is computed from named parameters, so a bolt, gear or spring is a function of its dimensions — change a value and the model reflects it. It is NOT a scanned solid model of the exact OEM part (those are ITAR-controlled); it's a dimensionally-grounded representative built from the cited FLIS data, paired with the authoritative Manual illustration tab for the real exploded view.

VERIFIED

All 13 families build valid meshes with the new params; gear teeth (12 vs 40) and spring turns (4 vs 20) change the geometry; backward-compatible with no extra params. Client-only, ES5 + WebGL with SVG fallback (RPS-safe).